

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Mapping and Monitoring Complex Chemical Components in Ambient Air using Fast GC/MS and Multivariate Data Analysis

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ABSTRACT

Considering the fact that the precise composition of many ambient environmental (air, water, soil) or industrial-process related samples represents a snapshot of the interaction of multiple chemical and physical reactions and transport phenomena, not unlike those encountered in a wide variety of molecular separation techniques, the ability of spatially and/or temporally distributed sample series to add new analytical dimensions should not come as a great surprise. From a multivariate analysis point of view one can simply look at the mass spectra or gas chromatograms of complex mixtures as the more or less linear sums of spectra or chromatograms representing individual mixture components. As long as one has a large enough set (i.e. the number of spectra or chromatograms should exceed the number of individually varying compounds or compound suites) of spectra or chromatograms in which the relative contributions of the target compounds or compound suites vary substantially it then will often be possible to numerically extract the profiles of the individual mixture components and determine their relative concentrations [Windig et al.; *Chemometrics and intelligent Systems*, 1, 151-165, 1987]. Although the procedure is simplified and quantitative analysis is more readily possible if the spectra or chromatograms of the individual mixture components are known, this is by no means an immutable requirement. This argument applies irrespective of whether one happens to be dealing with series of mass spectra representing two or more incompletely separated chromatographic peaks or with series of spectra (or chromatograms) representing fluctuating mixture concentrations in the field. Besides the relative ease with which temporally- or spatially-resolved measurement dimensions can be added when using fast, repetitive and/or distributed analysis modes, a particularly attractive characteristic of spatially and temporally distributed chemical and physical measurements in the field is that they often represent processes and trends with which the operator is already quite familiar, e.g. natural pollutant dispersion processes or industrial process trends. This helps make the novel information provided by these ancient analytical dimensions intuitively meaningful and transparent to an operator with limited training in analytical chemistry. Finally, it is well recognized that the inherently poor transmission of optical, microwave and radio signals in U/W environments severely limits the use of the kind of stand-off spectroscopic techniques for rapid area mapping which have been so spectacularly successful in astronomy and space science. Consequently, the new generation of fast, miniaturized, multidimensional point detectors (including a wide range of MS-based techniques), promises to provide a powerful alternative, whether deployed as fixed-grid distributed sensor systems or on fast roving (robotic and/or remotely controlled) platforms.